

A Distributed Delivery Architecture for User Generated Content Live Streaming over HTTP

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Agenda

- Challenges of user generated content
- Problem definition
- Proposed approach
- Results
- Conclusion and future work
- Q&A

User Generated Content

- With the advancement of mobile phones, live User Generated Content (UGC) has become very popular in today's video streaming applications, in particular with gaming and e-sport.
- Majority of UGC services use HTTP-based adaptive streaming (**HAS**) for video distribution.
- **UGC challenges:**
 - Technical **complexity** of managing **parallel** streams,
 - Difficult trade-offs with video **quality** and **latency**

Problem Definition

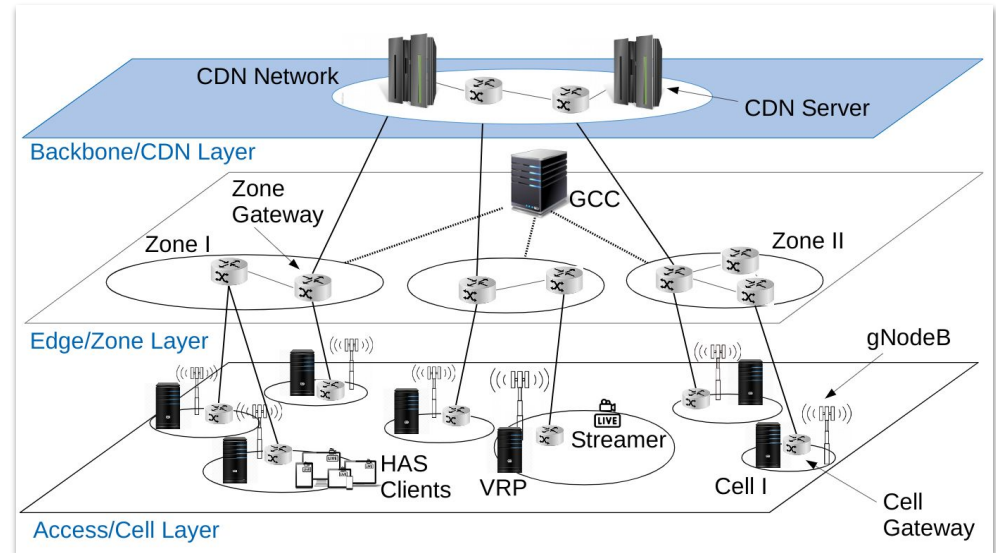
How to dynamically optimize **quality** and **latency** **fairness** for concurrent heterogeneous clients that rapidly join or leave various live UGC stream channels located in different geographical zones?

How to design a **scalable** solution?

QuaLA: Joint Quality-Latency Architecture

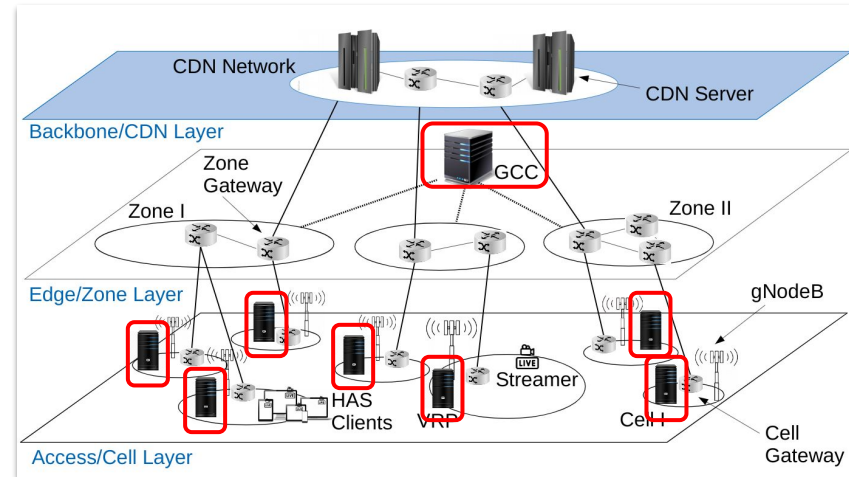
A **three-layer, fully distributed** architecture for HAS-enabled live UGC delivery over Internet

- Access/Cell Layer
- Edge/Zone Layer
- Backbone/CDN layer



QuaLA Design

- Formulate the joint quality-latency fairness problem for optimized QoE of live UGC streaming
 - Centralized QuaLA (C-QuaLA)
 - Distributed QuaLA (D-QuaLA)
- Introduce two concept of QoE fairness :
 - Fine-grained fairness
 - Coarse-grained fairness



Centralized QuaLA (C-QuaLA)

- Quality Constraints

Which qualities should be served by which CDN server?

(C1) $1 \leq \sum_{l \in L_{c,v}} \ell_{v,l}^{c,d} \leq |L_{c,v}|, \quad \forall c \in C, v \in V_c, d = \mathcal{S}(v)$

Binary variable

Set of Cells

Requested live channel by c

Source CDN server for v

The volume of transmitted data from source CDN servers to cells:

(C2) $\sum_{l \in L_{c,v}} (B_{v,l} - \ell_{v,l}^{c,d} Q_{v,l}) \leq (1 - f_{c,v}) \sum_{l \in L_{c,v}} B_{v,l},$

percentage of all qualities of channel v transmitted to cell c

Centralized QuaLA (C-QuaLA)

- Latency Constraints**

Satisfying the given latency by selecting optimal bitrates from CDN server to the requesting cell

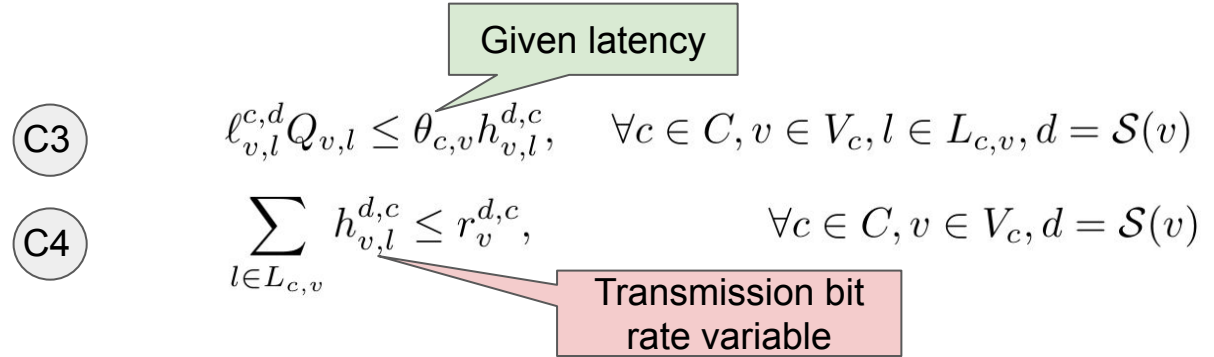


Diagram illustrating constraints C3 and C4 with callouts:

- C3** $\ell_{v,l}^{c,d} Q_{v,l} \leq \theta_{c,v} h_{v,l}^{d,c}, \quad \forall c \in C, v \in V_c, l \in L_{c,v}, d = \mathcal{S}(v)$ (Callout: Given latency)
- C4** $\sum_{l \in L_{c,v}} h_{v,l}^{d,c} \leq r_v^{d,c}, \quad \forall c \in C, v \in V_c, d = \mathcal{S}(v)$ (Callout: Transmission bit rate variable)

and adjusting bitrates from source cell to the CDN server

- C5** $h_{v,l}^{d,c} \leq H_{v,l}^d, \quad \forall c \in C, v \in V_c, l \in L_{c,v}, d = \mathcal{S}(v),$
- C6** $\sum_{l \in L_v} H_{v,l}^d \leq r_v^{x,d}, \quad \forall x \in C, v \in V_x^*, d = \mathcal{S}(v),$

Centralized QuaLA (C-QuaLA) ...

- **Resource Constraints:**

Considering available bandwidth of cell c , zone z , and CDN server d

$$\text{C7} \quad \sum_{v \in V_c} r_v^{d,c} + \sum_{v \in V_c^*} r_v^{c,d} \leq \omega_c, \quad \forall c \in C, d = \mathcal{S}(v)$$

$$\text{C8} \quad \sum_{c \in \mathcal{C}_z} \sum_{v \in V_c} r_v^{d,c} + \sum_{c \in \mathcal{C}_z} \sum_{v \in V_c^*} r_v^{c,d} \leq \Omega_z, \quad \forall z \in Z, d = \mathcal{S}(v)$$

$$\text{C9} \quad \sum_{c \in C} \sum_{v \in V_c} r_v^{d,c} + \sum_{c \in C} \sum_{v \in V_c^*} r_v^{c,d} \leq w_d \quad \forall d \in D \& d = \mathcal{S}(v)$$

Fine-grained C-QuaLA and Coarse-grained C-QuaLA

Providing quality-latency fairness for each live channel independently

Fine-grained Fairness

$$\begin{aligned}
 & \text{Maximize} \quad \boxed{\sum_{c \in C} \sum_{v \in V_c} F_v} + \sum_{c \in C} \sum_{v \in V_c} \sum_{l \in L_{c,v}} \rho_{v,l} \ell_{v,l}^{c,d} \\
 & \text{s.t. :} \quad \text{Constraints Eq.(1)–Eq.(9)} \\
 & \quad f_{c,v} \geq F_v \quad \forall c \in C, v \in V_c \\
 & \text{vars. :} \quad \ell_{v,l}^{c,d} \in \{0, 1\}, r_v^{c,d}, h_{v,l}^{c,d}, H_{v,l}^d, f_{c,v}, F_v \geq 0
 \end{aligned}$$

Minimum fairness of each live channel v

The fairness is determined over all live channels

Coarse-grained Fairness

$$\begin{aligned}
 & \text{Maximize} \quad \boxed{F} + \sum_{c \in C} \sum_{v \in V_c} \sum_{l \in L_{c,v}} \rho_{v,l} \ell_{v,l}^{c,d} \\
 & \text{s.t. :} \quad \text{Constraints Eq.(1)–Eq.(9)} \\
 & \quad f_{c,v} \geq F \quad \forall c \in C, v \in V_c \\
 & \text{vars. :} \quad \ell_{v,l}^{c,d} \in \{0, 1\}, r_v^{c,d}, h_{v,l}^{c,d}, H_{v,l}^d, f_{c,v}, F \geq 0
 \end{aligned}$$

Minimum fairness of all live channels

Distributed QuaLA(D-QuaLA)

- Mathematical distribution techniques:
 - Two blocks of functions and variables (ADMM: The alternating direction method of multipliers)[1]
 - Multi-block approaches:
 - Variable Splitting ADMM [2]
 - ADMM with Gaussian Back Substitution [3]
 - Proximal Jacobian ADMM (ProxJ-ADMM) [4]
 - Fast convergence rate

[1] S. Boyd, N. Parikh, and E. Chu, Distributed optimization and statistical learning via the alternating direction method of multipliers. Now Publishers Inc, 2011.

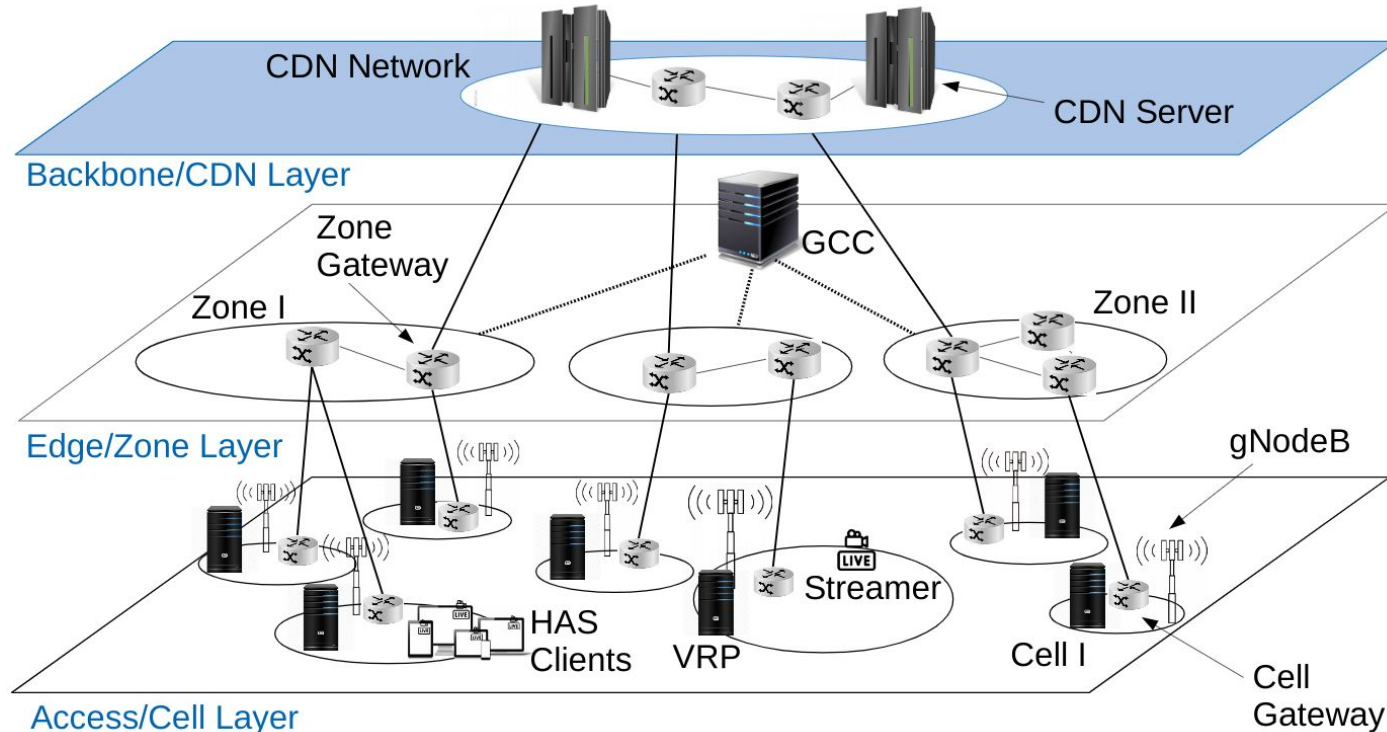
[2] D. P. Bertsekas and J. N. Tsitsiklis, Parallel and distributed computation: numerical methods. Prentice hall Englewood Cliffs, NJ, 1989, vol. 23.

[3] B. He, M. Tao, and X. Yuan, “Alternating direction method with gaussian back substitution for separable convex programming,” SIAM Journal on Optimization, vol. 22, no. 2, pp. 313–340, 2012.

[4] W. Deng, M.-J. Lai, Z. Peng, and W. Yin, “Parallel multi-block admm with $\mathcal{O}(1/k)$ convergence,” Journal of Scientific Computing, vol. 71, no. 2, pp. 712–736, 2017.

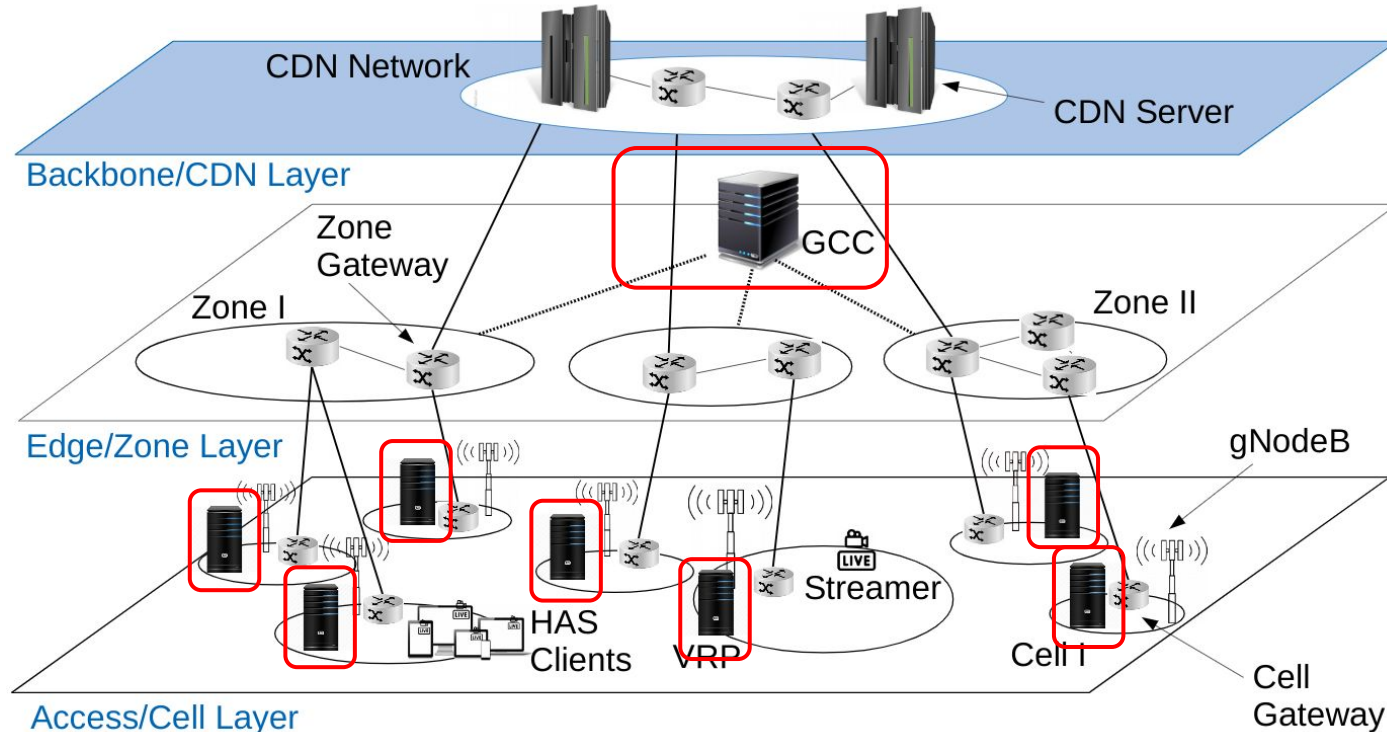
How Does Distributed QuaLA(D-QuaLA) Work?

Applying Proximal Jacobian ADMM (ProxJ-ADMM) method



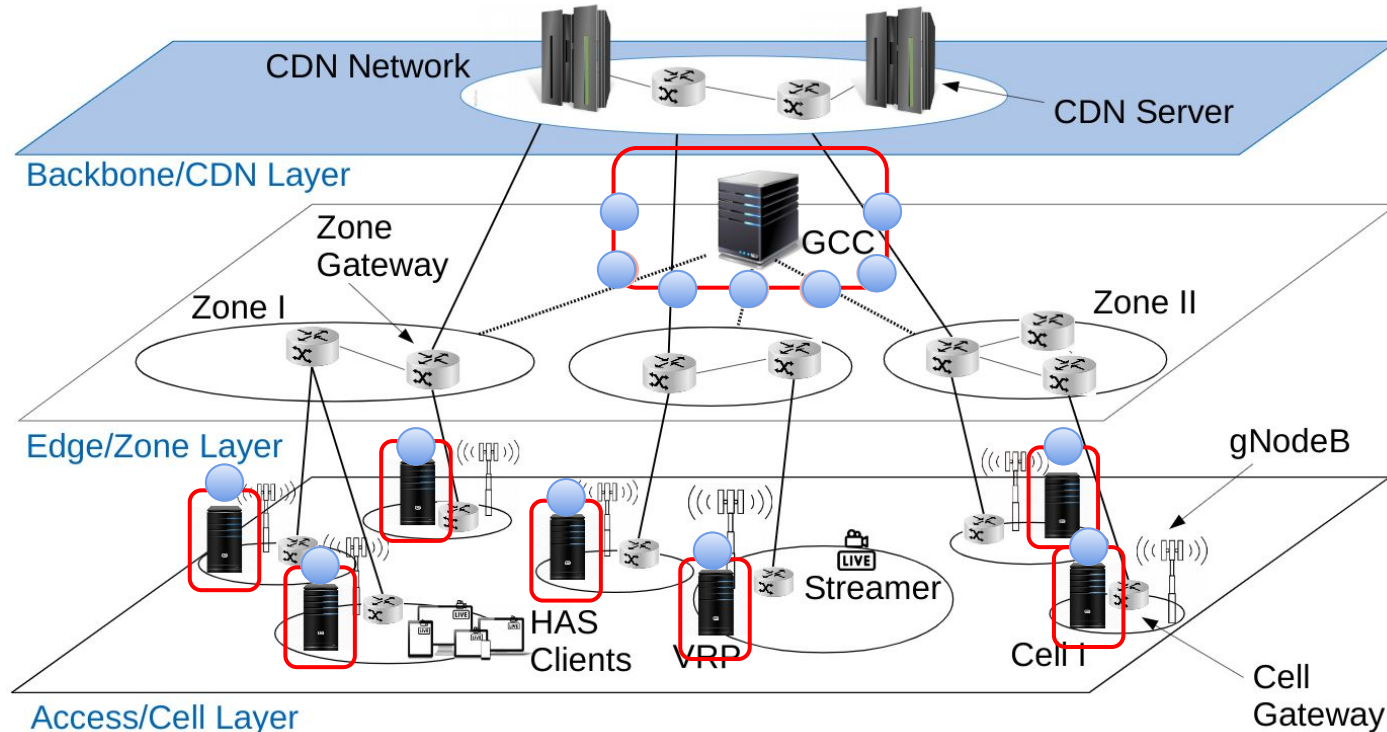
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Distributed QuaLA(D-QuaLA) for Coarse-grained Fairness

How to relax the binary variable $\ell_{v,l}^{c,d}$

1- Define the following two variables showing the outgoing and incoming video traffics to and from cells and CDN servers:

$x_{i,j}^v$ From CDN server j to cell i for video v

$y_{i,j}^v$ From cell i to CDN server j for video v

these two variables enable us to run the model over a group of cells in a distributed manner.

2- Define $B_{c,v} = \frac{\sum_{l \in L_{c,v}} Q_{v,l}}{\Theta_{,,}}$ as the required bitrate for downloading video v by cell c with respect to desired qualities

Distributed QuaLA(D-QuaLA) ...

$$\text{Minimize } \sum_{c \in C} \sum_{v \in V_c} f_{c,v}$$

$$B_{c,v} - x_{c,d}^v \leq f_{c,v} B_{c,v},$$

$$x_{c,d}^v \leq y_{c',d}^v, \quad \forall c, c' \in C, v \in \{V_c \cap V_{c'}^*\}, d \in D$$

$$\sum_{v \in V_c} x_{c,d}^v + \sum_{v \in V_c^*} y_{c,d}^v \leq \omega_c, \quad \forall c \in C, d \in D$$

$$\sum_{c \in C_z} \sum_{v \in V_c} x_{d,c}^v + \sum_{c \in C_z} \sum_{v \in V_c^*} y_{c,d}^v \leq \Omega_z, \quad \forall z \in Z$$

$$\sum_{c \in C} \sum_{v \in V_c} x_{d,c}^v + \sum_{c \in C} \sum_{v \in V_c^*} y_{d,c}^v \leq w_d, \quad \forall d \in D$$

$$x_{c,d}^v = y_{d,c}^v, \quad \forall c \in C, v \in V_c, d \in D, \text{ if } d = \mathcal{S}(v)$$

$$f_{c,v} = f_{c',v}, \quad \forall c, c' \in C, v \in \{V_c \cap V_{c'}\}$$

$$\text{var. } x_{c,d}^v, y_{d,c}^v, f_{c,v} \geq 0$$

Distributed QuaLA(D-QuaLA)...

By considering the equality constraints in the proposed model and having the Lagrangian multipliers λ and γ as well as the penalty parameters ρ and ρ , the Lagrangian equation can be represented as follows:

$$\begin{aligned}\mathcal{L}_{\rho_1, \rho_2}(f, x, y, \lambda, \gamma) = & \sum_{c \in C} \sum_{v \in V_c} f_{c,v} - \\ & \sum_{c \in C} \sum_{v \in V_c} \lambda(x_{c,d}^v - y_{d,c}^v) + \frac{\rho_1}{2} \sum_{c \in C} \sum_{v \in V_c} \|x_{c,d}^v - y_{d,c}^v\|_2^2 - \\ & \sum_{c \in C} \sum_{c' \in C} \sum_{v \in A} \gamma(f_{c,v} - f_{c',v}) + \frac{\rho_2}{2} \sum_{c \in C} \sum_{c' \in C} \sum_{v \in A} \|f_{c,v} - f_{c',v}\|_2^2\end{aligned}$$

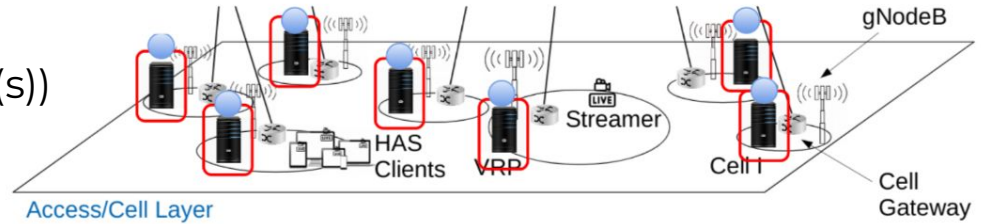
Distributed QuaLA(D-QuaLA)...

- Cell updates

$\mathbf{x}_c^{(k+1)}$ (if cell c has request(s))

$\mathbf{y}_c^{(k+1)}$ (if cell c hosts streamer(s))

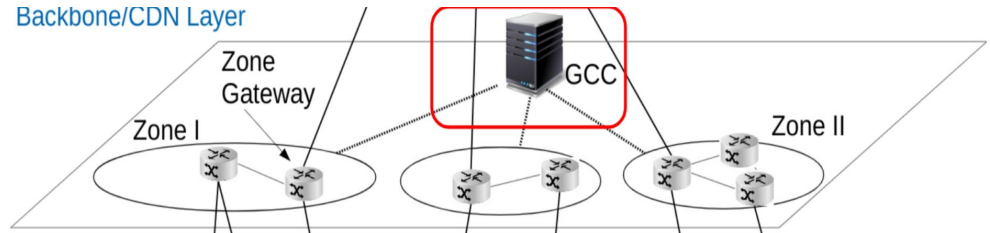
$f_{c,v}^{(k+1)}$ (if cell c has request(s))



- GCC updates the Lagrangian multipliers λ and γ as follows

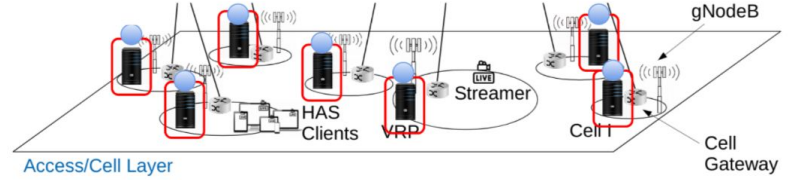
$\lambda^{(k+1)}$

$\gamma^{(k+1)}$



Distributed QuaLA(D-QuaLA)...

- Cell updates (if cell c hosts streamer(s))



$$\mathbf{y}_c^{(k+1)} =$$

$$\underset{\mathbf{y}_c}{\operatorname{argmin}} \left(\frac{\rho_1}{2} \sum_{v \in V_c^*} \|y_{c,d}^v - \hat{\lambda}_{d,c}^{v(k)}\|_2^2 + \frac{1}{2} \sum_{v \in V_c^*} \|y_{c,d}^v - y_{c,d}^{v(k)}\|_{P_y}^2 \right)$$

$$s.t. \quad B_{c,v} - x_{c,d}^v \leq f_{c,v} B_{c,v},$$

$$x_{c,d}^v \leq y_{c',d}^v, \quad \forall c' \in C, v \in \{V_c \cap V_{c'}^*\}, d \in D$$

$$\sum_{v \in V_c} x_{c,d}^v + \sum_{v \in V_c^*} y_{c,d}^v \leq \omega_c, \quad \forall d \in D$$

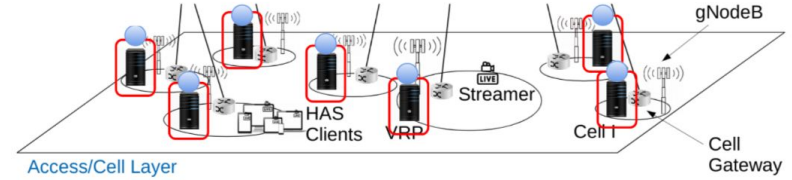
$$\sum_{c \in \mathcal{C}_z} \sum_{v \in V_c} x_{d,c}^v + \sum_{c \in \mathcal{C}_z} \sum_{v \in V_c^*} y_{c,d}^v \leq \Omega_z, \quad \forall z \in Z$$

$$\sum_{c \in C} \sum_{v \in V_c} x_{d,c}^v + \sum_{c \in C} \sum_{v \in V_c^*} y_{d,c}^v \leq w_d, \quad \forall d \in D$$

where $\hat{\lambda}_{d,c}^{v(k)} = x_{d,c}^{v(k)} + \frac{\lambda^{(k)}}{\rho_1}$

Distributed QuaLA(D-QuaLA)...

- Cell updates (if cell c has request(s))



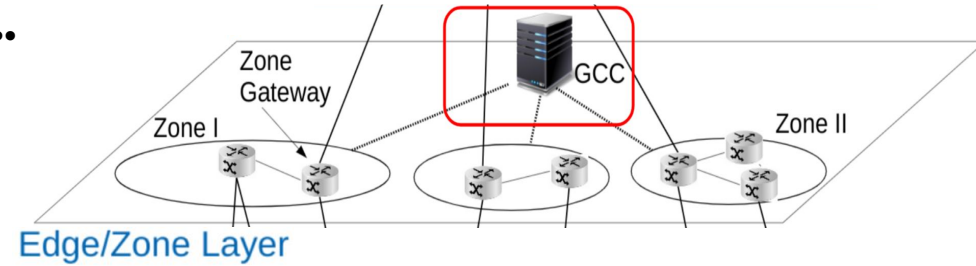
$$f_{c,v}^{(k+1)} =$$

$$\underset{f_{c,v}}{\operatorname{argmin}} \left(f_{c,v}^2 + \frac{\rho_2}{2} \|f_{c,v} - \bar{\gamma}_{c,v}^{(k)}\|_2^2 + \frac{1}{2} \|f_{c,v} - f_{c,v}^{(k)}\|_{P_f}^2 \right)$$

$$s.t. \quad B_{c,v} - x_{c,d}^v \leq f_{c,v} B_{c,v}, \quad \forall d \in D, \text{ if } d = \mathcal{S}(v)$$

$$\text{where } \bar{\gamma}_{c,v}^{(k)} = f_{c,v}^{(k)} + \frac{\gamma^{(k)}}{\rho_2}$$

Distributed QuaLA(D-QuaLA)...



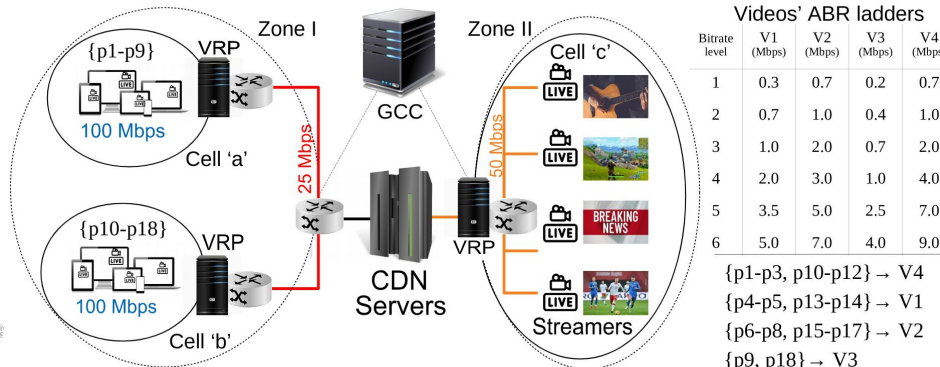
- GCC updates the Lagrangian multipliers λ and γ as follows

$$\lambda^{(k+1)} = \lambda^{(k)} - \nu \rho_1 \sum_{c \in C} \sum_{v \in V_c} \sum_{d \in D} x_{c,d}^{v(k+1)} - y_{d,c}^{v(k+1)},$$

$$\gamma^{(k+1)} = \gamma^{(k)} - \nu \rho_2 \sum_{c \in C} \sum_{c' \in C} \sum_{v \in A} f_{c,v}^{(k+1)} - f_{c',v}^{(k+1)},$$

Performance Evaluation

- To test the practicality of QuaLA in real-world scenarios, we implemented the following entities within CloudLab:
 - streamers use our video dataset to generate video sources (V1–V4)
 - viewers use the open-source Python-based HAS player (AStream) [34]
 - FFmpeg encoder with HAS packager (DASH-based)
 - Apache HTTP server as an origin
 - Python-based HTTP servers for cell VRPs and the GCC



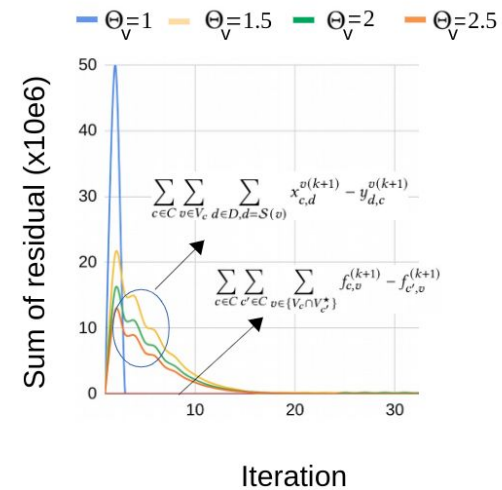
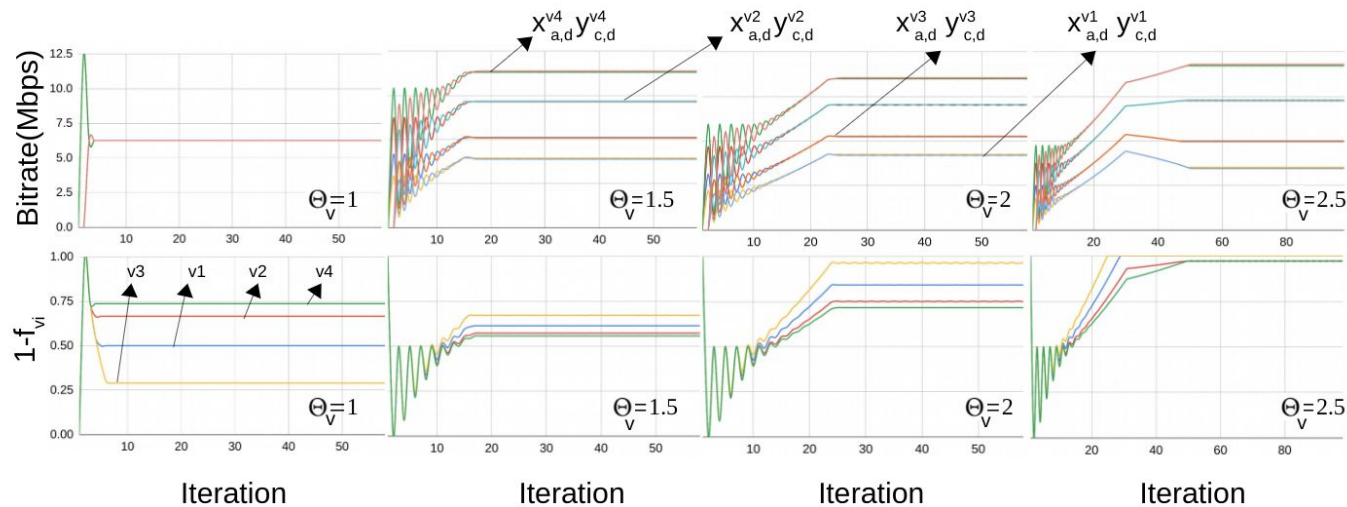
M1: all bitrates have the same priority

M2: the three lowest bitrates have the higher priority

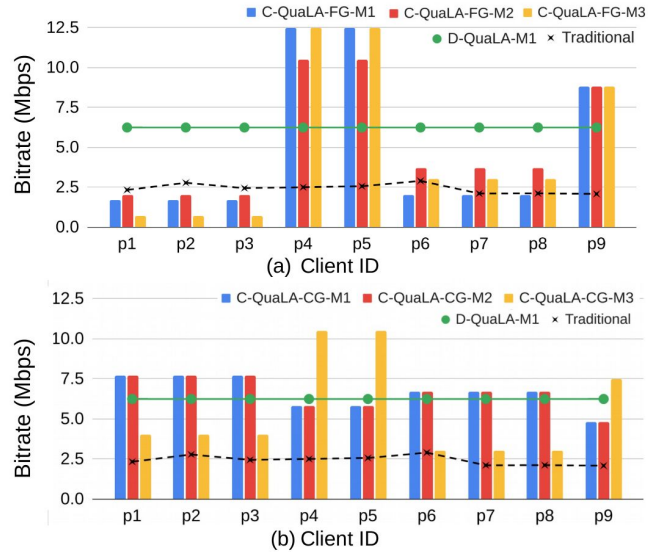
M3: the three highest bitrates have the higher priority

Performance of D-QualA

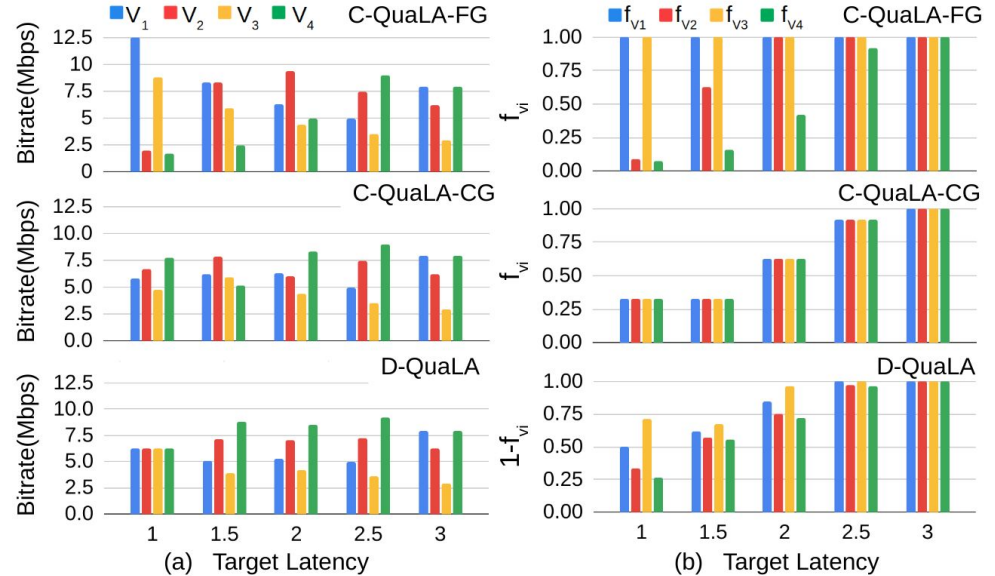
- Convergence time
- Sum of residual for different values of target latency



Comparison of C-QuaLA, D-QuaLA

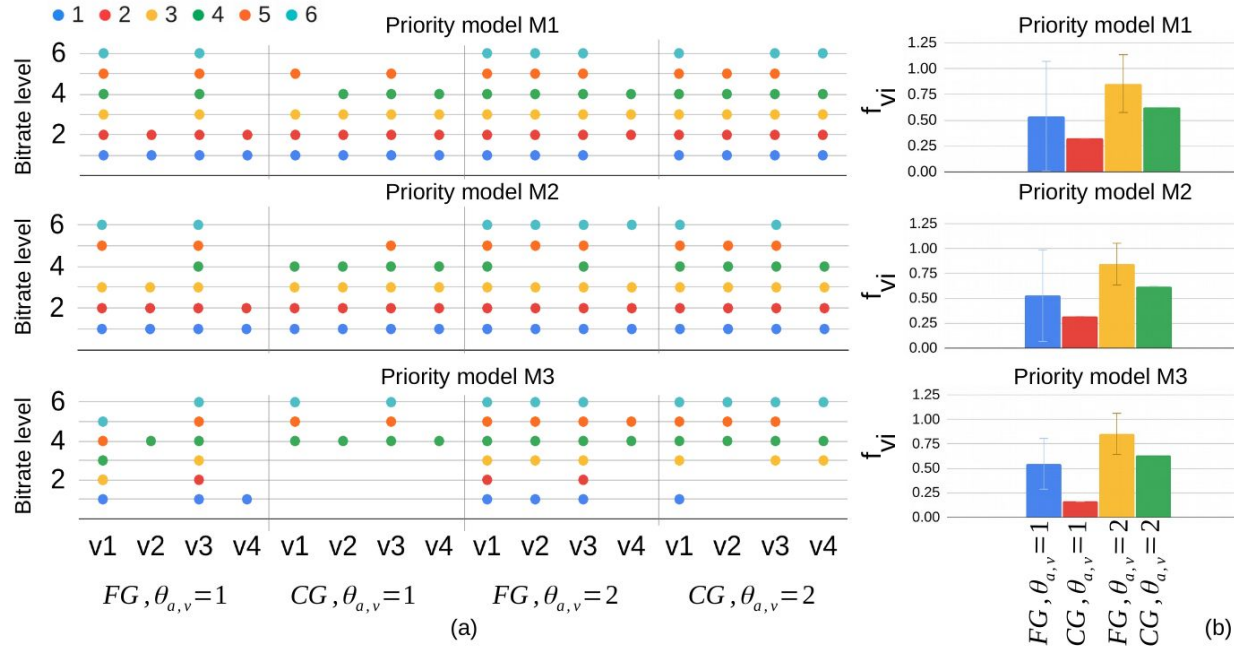


Comparison of C-QuaLA, D-QuaLA, and traditional approach in terms of serving bitrate



Performance of C-QuaLA and D-QuaLA for different values of target latency

Impact of priority on bitrate selection by C-QuaLA



Conclusion and Future Work

- we proposed QuaLA, a video streaming architecture that considers the high quality and low latency requirements of user-generated live content (UGC) applications.
 - Centralized- and Distributed-QuaLA
 - Fine-/coarse-grained fairness
- QuaLA showed 57% improvement in bitrate and fairness at various given target latency values among all the clients.
- Extend QuaLA to support learning-based approach in the future
- Design D-QuaLA for the fine-grained fairness model

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